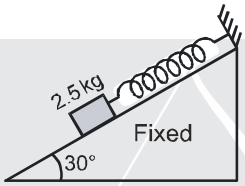
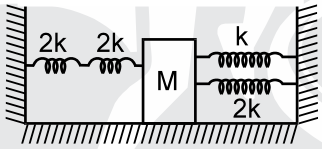
SYLLABUS : SIMPLE HARMONIC MOTION

1. A smooth inclined plane having angle of inclination 30° with horizontal has a mass 2.5 kg held by a spring which is fixed at the upper end as shown in figure. If the mass is taken 2.5 cm up along the surface of the inclined plane, the tension in the spring reduces to zero. If the mass is then released, the angular frequency of oscillation in radian per second is



(A) 0.707 (B) 7.07 (C) 1.414 (D) 14.14

2. Four massless springs whose force constants are 2k, 2k, k and 2k respectively are attached to a mass M kept on a frictionless plane (as shown in figure). If the mass M is displaced in the horizontal direction, then the frequency of the system.



(A) $\frac{1}{2\pi}\sqrt{\frac{k}{4M}}$ (B) $\frac{1}{2\pi}\sqrt{\frac{4k}{M}}$ (C) $\frac{1}{2\pi}\sqrt{\frac{k}{7M}}$ (D) $\frac{1}{2\pi}\sqrt{\frac{7k}{M}}$

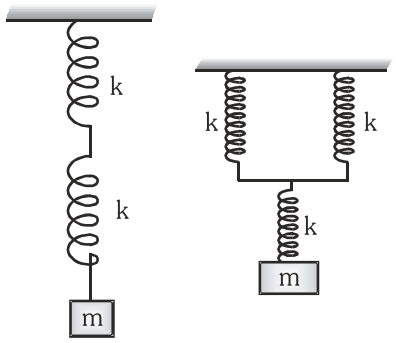
3. A heavy brass-sphere is hung from a spiral spring and it executes vertical vibrations with period T. The ball is now immersed in nonviscous liquid with a density one-tenth that of brass. When set into vertical vibrations with the sphere remaining inside the liquid all the time, the period will be-

(A) $\left[\frac{9}{10}\right]T$ (B) $T\sqrt{\left(\frac{10}{9}\right)}$ (C) Unchanged (D) $T\sqrt{\left(\frac{9}{10}\right)}$

4. The total mechanical energy of a particle of mass m executing SHM with the help of a spring is $E = (1/2)m\omega^2A^2$. If the particle is replaced by another particle of mass m/2 while the amplitude A remains same. New mechanical energy will be :

(A) $\sqrt{2}$ E (B) 2E (D) E/2 (D) E

5. Some springs are combined in series and parallel arrangement as shown in the figure and a mass m is suspended from them. The ratio of their frequencies will be :–



(A) 1 : 1 (B) 2 : 1 (C) $\sqrt{3}$: 2 (D) 4 : 1

6. A spring-mass system oscillates in a car. If the car accelerates with a constant acceleration on a horizontal road, the frequency of oscillation will

(A) increase (B) decrease (C) remain same (D) become zero.

7. A horizontal spring is connected to a mass M. It executes simple harmonic motion. When the mass M passes through its mean position, an object of mass m is put on it and the two move together. The ratio of frequencies before and after will be-

(A) $\left(1+\frac{m}{M}\right)^{1/2}$ (B) $\left(1+\frac{m}{M}\right)$ (C) $\left(\frac{M}{M+m}\right)^{1/2}$ (D) $\left(\frac{M}{M+m}\right)$

8. A spring mass a system has a time period of 2 second. What should be the spring constant of spring if the mass of the block is 10 grams ?

(A) 0.1 N/m (B) 0.2 N/m (C) 1 N/m (D) 2.1 N/m

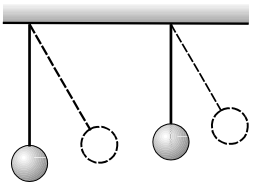
9. A pendulum is suspended in a lift and its period of oscillation when the lift is stationary is T_0. What must be the acceleration of the lift for the period of oscillation of the pendulum to be $T_0/2$?

(A) 2g downward (B) 2g upward (C) 3g downward (D) 3g upward

10. A pendulum clock that keeps correct time on the earth is taken to the moon. It will run

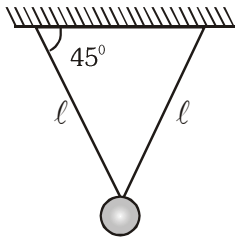
(A) at correct rate (B) 6 times faster (C) $\sqrt{6}$ times faster (D) $\sqrt{6}$ times slower

11. Two pendulums at rest start swinging together. Their lengths are respectively 1.44 m and 1 m. They will again start swinging in same phase together after (of longer pendulum) :



(A) 1 vibration (B) 3 vibrations (C) 4 vibrations (D) 5 vibrations

12. Time period of small oscillation (in a vertical plane normal to the plane of strings) of the bob in the arrangement shown will be



(A) $2\pi\sqrt{\frac{\ell}{g}}$ (B) $2\pi\sqrt{\frac{\ell}{\sqrt{2}g}}$ (C) $2\pi\sqrt{\frac{\sqrt{2}\ell}{g}}$ (D) $2\pi\sqrt{\frac{2\ell}{g}}$

13. A simple pendulum has some time period T. What will be the percentage change in its time period if its amplitude is decreased by 5%?

(A) 6 % (B) 3 % (C) 1.5 % (D) 0 %

14. A wall clock uses simple pendulum. It is accurate at earth surface .If it is taken to high altitude:

(A) it will run fast

(B) its length should be increased to keep it accurate.

(C) its length should be decreased to keep it accurate.

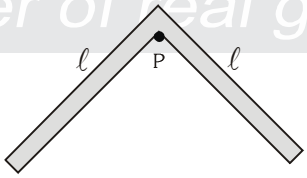
(D) Even if length is changed it cannot be accurate.

15. Find the length of seconds pendulum at a place where $g = \pi^2$ m/s^2.

(A) 1 m (B) 2 m (C) 3 m (D) 5 m

16. A system of two identical rods (L-shaped) of mass m and length ℓ are resting on a peg P as shown in the figure. If the system is displaced in its plane by a small angle θ, find the period of oscillations

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(A) $2\pi\sqrt{\frac{\sqrt{2}\ell}{3g}}$ (B) $2\pi\sqrt{\frac{2\sqrt{2}\ell}{3g}}$ (C) $2\pi\sqrt{\frac{2\ell}{3g}}$ (D) $3\pi\sqrt{\frac{\ell}{3g}}$

17. A 25 kg uniform solid sphere with a 20 cm radius is suspended by a vertical wire such that the point of suspension is vertically above the centre of the sphere. A torque of 0.10 N-m is required to rotate the sphere through an angle of 1.0 rad and then the orientation is maintained. If the sphere is then released, its time period of the oscillation will be :

(A) π second (B) $\sqrt{2}\pi$ second (C) 2π second (D) 4π second

18. The distance between the point of suspension and the centre of gravity of a compound pendulum is ℓ and the radius of gyration about the horizontal axis through the centre of gravity is k, then its time period will be

(A) $2\pi\sqrt{\frac{\ell+k}{g}}$ (B) $2\pi\sqrt{\frac{\ell^2+k^2}{\ell g}}$ (C) $2\pi\sqrt{\frac{\ell+k^2}{g}}$ (D) $2\pi\sqrt{\frac{2k}{\ell g}}$

19. An oscillation is described by the equation $x=A \sin 2\pi\gamma_1 t$ where A changes with time according to the law $A= A_0 (1+\cos 2\pi\gamma_2 t)$ where A_0 is constant. Find the ratio of frequencies of harmonic oscillations forming oscillation

(A) $\gamma_1 : \gamma_2 (\gamma_1 - \gamma_2)$ (B) $\gamma_1 : (\gamma_1 - \gamma_2):(\gamma_1 + \gamma_2)$

(C) $\gamma_1 : \gamma_2 : (\gamma_2 - \gamma_1)$ (D) $\gamma_1 : \gamma_2 (\gamma_1 + \gamma_2)$

20. When two mutually perpendicular simple harmonic motions of same frequency, amplitude and phase are superimposed

(A) the resulting motion is uniform circular motion.

(B) the resulting motion is a linear simple harmonic motion along a straight line inclined equally to the straight lines of motion of component ones.

(C) the resulting motion is an elliptical motion, symmetrical about the lines of motion of the components.

(D) the two S.H.M. will cancel each other.

21. A simple harmonic motion is given by $y = 5 (\sin 3\pi t + \sqrt{3} \cos 3\pi t)$. What is the amplitude of motion if y is in m ?

(A) 100 cm (B) 5 m (C) 200 cm (D) 1000 cm

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22. The position vector of a particle moving in x-y plane is given by

$\vec{r} = (A \sin\omega t)\hat{i} + (A \cos\omega t)\hat{j}$ then motion of the particle is :

(A) SHM (B) on a circle

(C) on a straight line (D) with constant acceleration

23. In a simple harmonic oscillator, at the mean position:

(A) kinetic energy is minimum, potential energy is maximum

(B) both kinetic and potential energies are maximum

(C) kinetic energy is maximum, potential energy is minimum

(D) both kinetic and potential energies are minimum

24. The length of a simple pendulum executing simple harmonic motion is increased by 21%. The percentage increase in the time period of the pendulum of increased length is :

(A) 11% (B) 21% (C) 42% (D) 10.5%

25. A body executes simple harmonic motion. The potential energy (P.E.), the kinetic energy (K.E.) and total energy (T.E.) are measured as function of displacement x. Which of the following statements is true?

(A) K.E. is maximum when x = 0 (B) T.E. is zero when x = 0

(C) K.E. is maximum when x is maximum (D) P.E. is maximum when x = 0

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ANSWER KEY

1.	(D)	**2.**	(B)	**3.**	(C)	**4.**	(D)	**5.**	(C)
6.	(C)	**7.**	(A)	**8.**	(A)	**9.**	(D)	**10.**	(D)
11.	(D)	**12.**	(B)	**13.**	(D)	**14.**	(C)	**15.**	(A)
16.	(B)	**17.**	(D)	**18.**	(B)	**19.**	(B)	**20.**	(B)
21.	(D)	**22.**	(B)	**23.**	(C)	**24.**	(D)	**25.**	(A)